

# FELIX SHAW - BELL

(+64) 221-277-204 • fshawbell@gmail.com • fshawbell.github.io

## PROFILE

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- Citizenship: New Zealand
- Current residence: Auckland, New Zealand

A high-achieving and motivated mathematician with aspirations of making a lasting contribution to both mathematics and the wider community. I focus my mathematical abilities on real-world modelling, particularly in problems involving uncertainty. My strong academic record is complemented by industry experience, allowing me to apply myself in both a rigorous and practical manner.

## EXPERIENCE

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### University of Auckland

March 2026 - June 2026

*Graduate Teaching Assistant (part-time)*

*Auckland, New Zealand*

- Delivered tutorials to classes of 20 to 40 students for first-year mathematics courses.

### Shift Finance

February 2025 - August 2025

*Data Scientist (full-time)*

*Sydney, Australia (Remote)*

- Developed a machine learning model to assist and automate credit risk assessments.
- Consolidated multiple data sources into SQL tables, now widely used within the company.

### University of Auckland

November 2024 - February 2025

*Mathematics Research Scholar (full-time)*

*Auckland, New Zealand*

- Worked under Professor Simon Harris and Dr Jesse Goodman to study branching processes.
- Funded by the University of Auckland Summer Research Scholarship.

### Shift Finance

November 2023 - November 2024

*Data Analyst Intern (full-time & part-time)*

*Sydney, Australia (On-site & Remote)*

- Built and deployed a company-specific Python library providing a machine learning pipeline.
- Contributed to the development of Shift's credit risk models.
- Carried out feature engineering work to enhance model performance from existing datasets.

## EDUCATION

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### University of Auckland

July 2025 - July 2026

*MSc, Applied Mathematics (awaiting results)*

*Auckland, New Zealand*

- Key areas of study: stochastic modelling, tipping point systems, approximation methods.
- Masters dissertation: '*Eigenmode approximations of noisy tipping-point systems*', supervised by Professor Graham Donovan. The project applied eigenmode approximations of the Fokker-Planck equation to estimate the risk of tipping in tipping point systems.

### University of Auckland

February 2021 - November 2024

*BSc (Hons), Applied Mathematics, First Class*

*Auckland, New Zealand*

- Key areas of study: mathematical modelling, numerical methods, machine learning.
- Honours dissertation: '*Stochastic modelling of soil nitrogen in intensive agriculture*', supervised by Dr Steve Taylor. The project developed a system of stochastic differential equations to model the important issue of nitrogen pollution into agricultural waterways.

## SKILLS

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- Technical: MATLAB, Python, SQL,  $\text{\LaTeX}$ .
- Languages: English (native), Spanish (elementary).

## CONFERENCES AND WORKSHOPS

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**New Zealand Mathematics and Statistics Postgraduate Conference**      November 2025  
*'Eigenmode approximations of noisy tipping-point systems'* (presentation)

## AWARDS AND HONOURS

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- University of Auckland Research Masters Scholarship (2025, stipend and free tuition).
- University of Auckland Summer Research Scholarship (2024, stipend).
- University of Auckland Young Scholars Program (2020).
- NZQA Calculus Scholarship (2020).
- NZQA Physics Scholarship (2020).
- Rotary Science & Technology Forum Sponsorship (2019).